

Topological Stiffness and Geometric Characterization of a 36-Node Kuramoto Ring

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Abstract

We present a numerical characterization of a 36-node Kuramoto ring with nearest-neighbor coupling and Gaussian frequency heterogeneity. Three distinct dynamical regimes are identified: incoherent ($K < 3\sigma_\omega$), chimera ($3 \leq K/\sigma_\omega \leq 5$), and synchronized ($K > 5\sigma_\omega$). The chimera regime is characterized by the coexistence of coherent and incoherent oscillator clusters and corresponds to a maximum of geometric heterogeneity as measured by the standard deviation of the discrete Ollivier-Ricci curvature. Spectral analysis of the weighted graph Laplacian shows that algebraic connectivity (Fiedler value λ_2) saturates near $K \approx 4.5\sigma_\omega$, while Dirichlet energy and Ricci curvature variance approach their asymptotic minima near $K \approx 20\text{--}22\sigma_\omega$. These three independent geometric measures provide a consistent picture: the ring transitions from a geometrically fragmented state to a topologically stiff, uniformly curved manifold. Parallels to biological coupled-oscillator systems (suprachiasmatic nucleus, cardiac SA node, cortical gamma oscillations) are noted.

1. Introduction

The Kuramoto model of coupled phase oscillators has been extensively studied for all-to-all coupling topologies. Nearest-neighbor ring topologies introduce qualitatively different behavior, including chimera states — spatially coexistent coherent and incoherent clusters — first described by Kuramoto and Battogtokh (2002) and subsequently observed experimentally in chemical, optical, and neural systems.

The present work does not introduce new theoretical results. It provides a combined numerical characterization using three established geometric frameworks applied to the specific case $N = 36$. The combination of order-parameter dynamics, discrete Ollivier-Ricci curvature, and spectral Laplacian analysis gives a consistent geometric picture of the synchronization transition that has not been presented together for this topology and dimension.

2. Model

The system is defined by the nearest-neighbor Kuramoto equations on a ring of $N = 36$ nodes with periodic boundary conditions:

$$d\theta_i/dt = \omega_i + (K/2)[\sin(\theta_{i+1} - \theta_i) + \sin(\theta_{i-1} - \theta_i)]$$

Natural frequencies ω_i are drawn from a Gaussian distribution with zero mean and unit variance ($\sigma_\omega = 1$, normalized units). Coupling K is swept from 0.5 to $22\sigma_\omega$. Numerical integration uses RK45 with maximum step $0.05\sigma_\omega^{-1}$ and integration time $100\sigma_\omega^{-1}$.

3. Results

3.1 Three Dynamical Regimes

Numerical integration identifies three regimes by the stationary value of the global order parameter $r = |(1/N)\sum e^{i\theta_k}|$:

Regime	K range	r (stationary)	Character
Incoherent	$K < 3.69\sigma_\omega$	$r \approx 0.18$	Random phases
Chimera	$3.69\sigma_\omega < K < 20.31\sigma_\omega$	$0.2 < r < 0.7$	Bistable, hysteresis
Synchronous	$K > 20.31\sigma_\omega$	$r \approx 1$	Uniform phase locking

3.2 Ollivier-Ricci Curvature

The discrete Ollivier-Ricci curvature of each edge is computed via optimal transport of probability measures on 3-point neighborhoods ($\alpha = 0.5$). Key findings: In the chimera regime, maximum curvature heterogeneity occurs at $K \approx 3.69\sigma_\omega$ where $\sigma(\kappa)$ reaches its peak. Coherent clusters show $\kappa > 0$ (sphere-like); incoherent edges show $\kappa < 0$. At $K = 20.31\sigma_\omega$, variance is reduced by ~67% relative to the chimera peak.

3.3 Spectral Analysis of the Weighted Graph Laplacian

Edge weights $w_{i,i+1} = (K/2)|\cos(\theta_{i+1} - \theta_i)|$ define the linearized coupling strength at the stationary state. Three spectral measures:

Measure	$K=3.69\sigma_\omega$ (Chimera)	$K=20.31\sigma_\omega$ (High coupling)	Behavior
Fiedler λ_2	0.033	0.302	Monotone increasing
Ricci κ_{min} (MAX)	0.255	0.084	Decreasing, sat. ~22%
Dirichlet E	6.67	5.51	Decreasing, min ~18%

Algebraic connectivity saturates (inflection point) near $K \approx 4.5\sigma_\omega$. Full geometric unification (minimum $\sigma(\kappa)$, minimum $E_{Dirichlet}$) requires $K \approx 20-22\sigma_\omega$.

3.4 Biological Analogies

System	K/σ_ω	Regime (est.)	Functional role
SCN (circadian)	~1.9	Near-incoherent	Flexible entrainment
SA node (cardiac)	~3.5	Chimera boundary	Robust pacemaking
Cortical gamma	~3.2	Chimera	Information processing

* Biological K/σ values are order-of-magnitude estimates derived from literature parameter ranges, not direct measurements. They serve as contextual analogies on the universal synchronization curve, not quantitative predictions.

4. Discussion

The geometric characterization of the $N=36$ Kuramoto ring reveals that the chimera state has a precise geometric signature: maximum Ricci curvature heterogeneity, low algebraic connectivity, and high Dirichlet energy. The synchronized state corresponds to a topologically stiff, low-resistance, uniformly curved manifold.

Two characteristic scales emerge: $K \approx 4.5\sigma_\omega$ marks the transition to strong-expander behavior; $K \approx 20\text{--}22\sigma_\omega$ marks full geometric unification. These scales are consistent across all three geometric frameworks, suggesting a robust geometric interpretation of the chimera transition in nearest-neighbor rings.

5. Conclusions

1. The $N=36$ nearest-neighbor Kuramoto ring exhibits three dynamical regimes with distinct geometric signatures.
2. The chimera state corresponds to maximum Ollivier-Ricci curvature heterogeneity at $K \approx 3.69\sigma_\omega$.
3. Algebraic connectivity saturates near $K \approx 4.5\sigma_\omega$; full geometric stiffness requires $K \approx 20\text{--}22\sigma_\omega$.
4. Three independent geometric measures (Ricci variance, Fiedler value, Dirichlet energy) give a consistent picture of the synchronization transition.
5. Parallels to biological oscillator networks are noted as contextual analogies on the universal synchronization curve.

References

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